APARNA A

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SUMMARY

Electrical and Electronics Engineering student (graduating 2027) with a solid grounding in data structures, OOP and DBMS. Has built an ESP32-based battery monitoring system, a responsive e-commerce site and a GPS/GSM vehicle tracker, and is at her best when tracing a bug back to its root cause. Looking for a software engineering internship or entry-level role where she can keep building and shipping.

EDUCATION

B.E., Electrical and Electronics Engineering | Velalar College of Engineering and Technology, Erode — 2023 - 2027
CGPA: 8.05/10

Higher Secondary (Class XII): 81.6% | Thamarai Matric Higher Secondary School, Erode — 2023
Secondary School (Class X): 100% | Thamarai Matric Higher Secondary School, Erode — 2021

TECHNICAL SKILLS

Languages: Python, C, JavaScript
Web: HTML5, CSS3, JavaScript, Responsive Design
CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Data Modeling
Embedded & IoT: ESP32, Arduino, Embedded C, GPS/GSM Modules, Sensor Interfacing
Tools & Practices: Git/GitHub, VS Code, Arduino IDE, Debugging, Root-Cause Analysis, Technical Documentation

PROJECTS

Solar-Based Smart EV Charging Station with Battery Management System — May 2026
ESP32, Embedded C, IoT, Python

- Built a Battery Management System that tracks voltage, current, temperature and State of Charge in real time.
- Coded automated fault alerts that flag abnormal readings without anyone watching the system.
- Programmed the ESP32 control system across sensor, communication and display modules in a 4-member team; debugged issues found in testing.

E-Commerce Web Application — January 2026
HTML5, CSS3, JavaScript, OOP

- Built a responsive store with a product catalogue, shopping cart and admin dashboard, structured around OOP design.
- Designed the interface to work across phones and desktops, from browsing through to checkout.

GPS & GSM Based Bus Tracking System — November 2025
Arduino, GPS, GSM, C

- Built a real-time vehicle tracker that combines GPS location data with GSM transmission.
- Traced and fixed communication failures between hardware modules so data delivery stayed consistent.

EXPERIENCE

Python Trainee | Mastermind Techno Solutions — Oct 16 - 26, 2025

- Found and fixed errors across several Python applications by working through each to its root cause.
- Troubleshot code with mentors and peers, documented the fixes and confirmed they held up.

Student Trainee | Seshasayee Paper and Boards Ltd. — Jun 16 - 23, 2025

- Observed industrial workflows, quality control and issue tracking in a live plant environment.

ACHIEVEMENTS & CERTIFICATIONS

- Smart India Hackathon 2025: participated with "AI Train Scheduling for KMRL" (Sep 2025).
- Presented "IoT-Based Gravity Irrigation System" at Karpagam Institute of Technology (Feb 2026).
- Top 50 students in the College Placement Training Program.
- Deloitte Technology Virtual Job Simulation, Forage (Mar 2026): coding and problem-solving tasks.
- Python: Concurrent Programming & Multiprocessing, Skillsoft (Mar 2026).
- Basics of Python, Infosys Springboard (Mar 2026).